```python
#Problem 1 – The Gaming Tournament

scores = [520, 780, 430]

if scores[0] == scores[1] or scores[1] == scores[2] or scores[0] == scores[2]:

    print("Tie Detected")

else:

    first = second = third = scores[0]

    for score in scores:

        if score > first:

            third = second

            second = first

            first = score

        elif score > second and score != first:

            third = second

            second = score

        elif score > third and score != second and score != first:

            third = score

    print("First Rank:", first)

    print("Second Rank:", second)

    print("Third Rank:", third)


#Problem 2 – Pizza Party

orders = [

    "Veg",

    "Paneer",

    "Veg",

    "Cheese",

    "Paneer",

    "Veg"

]
```

```
unique = set(orders)
print("Unique Pizza Types:")
for pizza in unique:
    print(pizza)
print("Total Unique Pizzas:", len(unique))

#Problem 3 – ATM Machine
balance = 15000
withdraw = 7000
if withdraw < 0:
    print("Invalid Amount")
elif withdraw <= balance:
    balance = balance - withdraw
    print("Withdrawal Successful")
    print("Remaining Balance:", balance)
else:
    print("Insufficient Balance")

#Problem 4 – Scholarship Selection
marks = [96, 81, 55, 74, 67, 92]
for mark in marks:
    if mark >= 90:
        print(mark, "- Full Scholarship")
    elif mark >= 75:
        print(mark, "- Half Scholarship")
    elif mark >= 60:
        print(mark, "- Training Program")
    else:
        print(mark, "- Better Luck Next Time")
```

```
#Problem 5 – Traffic Signal Simulation
signal = "Green"
if signal == "Red":
    print("STOP")
elif signal == "Yellow":
    print("READY")
elif signal == "Green":
    print("GO")
else:
    print("Signal Error")

#Problem 6 – Shopping Cart
prices = [250, 180, 450, 120, 600]
total = 0
highest = prices[0]
lowest = prices[0]
for price in prices:
    total += price
    if price > highest:
        highest = price
    if price < lowest:
        lowest = price
print("Total Bill:", total)
print("Highest Price:", highest)
print("Lowest Price:", lowest)
```

```python
#Problem 7 – Movie Ticket Checker
age = 16
ticket = True
if age >= 18 and ticket == True:
    print("Entry Allowed")
else:
    print("Entry Denied")



#Problem 8 – Library System
books = [
    "Python",
    "Java",
    "Python",
    "C",
    "C++",
    "Java"
]
unique = set(books)
print("Unique Books:")
for book in unique:
    print(book)
print("Number of Unique Books:", len(unique))
if "Python" in books:
    print("Python Exists")
else:
    print("Python Not Found")
```

```
#Problem 9 – Race Elimination
players = [
    "Ajay",
    "Ravi",
    "Ali",
    "Kiran",
    "John"
]
for player in players:
    if player == "Ali":
        continue
    print(player)

#Problem 10 – Bus Capacity
passengers = []
for i in range(1, 51):
    passengers.append(i)
    if len(passengers) == 40:
        print("Bus Full")
        break

#Problem 11 – Lucky Number Hunt
numbers = [12, 8, 21, 15, 9, 18, 27]
for num in numbers:
    if num % 3 == 0 and num % 7 == 0:
        print("Lucky Number Found:", num)
        break
```

```python
#Problem 12 – Fruit Market
fruits = [
    "Apple",
    "Banana",
    "Mango",
    "Orange",
    "Apple",
    "Mango"
]
if "Mango" in fruits:
    print("Available")
else:
    print("Out of Stock")
unique = set(fruits)
print("Unique Fruits:")
for fruit in unique:
    print(fruit)

#Problem 13 – Guess the Secret Number
secret = 25
guesses = [10, 14, 25, 31, 42]
for guess in guesses:
    if guess == secret:
        print("Correct Guess")
        break
```

```python
#Problem 14 – Hospital Queue
patients = [
    "Rahul",
    "Emergency",
    "Sita",
    "Ramesh",
    "Akash"
]
for patient in patients:
    if patient == "Emergency":
        print("Treat Immediately")
        break
    print(patient)

#Problem 15 – Sports Academy
scores = [45, 82, 61, 93, 55, 78]
count = 0
for score in scores:
    if score >= 60:
        print(score, "- Qualified")
        count += 1
    else:
        print(score, "- Not Qualified")
print("Total Qualified:", count)
```

```python
#Problem 16 – Mobile Password Checker
correct_password = 1234
user_password = 1111
if user_password == correct_password:
    print("Phone Unlocked")
else:
    print("Wrong Password")

#Problem 17 – Cab Booking
fare = 850
if fare > 1000:
    discount = fare * 20 / 100
elif fare > 500:
    discount = fare * 10 / 100
else:
    discount = 0
final_fare = fare - discount
print("Final Fare:", final_fare)

#Problem 18 – School Attendance
attendance = [
    "Present",
    "Absent",
    "Present",
    "Present",
    "Absent"
]
present = 0
absent = 0
```

```
for status in attendance:
    if status == "Present":
        present += 1
    else:
        absent += 1
print("Total Present:", present)
print("Total Absent:", absent)

#Problem 19 – International Conference
countries = [
    "India",
    "USA",
    "India",
    "Japan",
    "Germany",
    "USA",
    "France"
]
unique = set(countries)
print("Unique Countries:")
for country in unique:
    print(country)
print("Total Countries:", len(unique))
```

#Problem 20 – Space Mission Control

```
systems = (
    "Engine",
    "Fuel",
    "Navigation",
    "Communication",
    "Cooling"
)
status = [
    "OK",
    "OK",
    "FAIL",
    "OK",
    "OK"
]
for i in range(len(systems)):
    print("Checking", systems[i], ":", status[i])
    if status[i] == "FAIL":
        print("Launch Aborted!")
        break
else:
    print("Ready for Launch")
```